

PHYSICS 9 | KINEMATICS

FROM POSITION GRAPH TO VELOCITY GRAPH

Use the exact Class 4-1 motion to construct the matching velocity-time graph

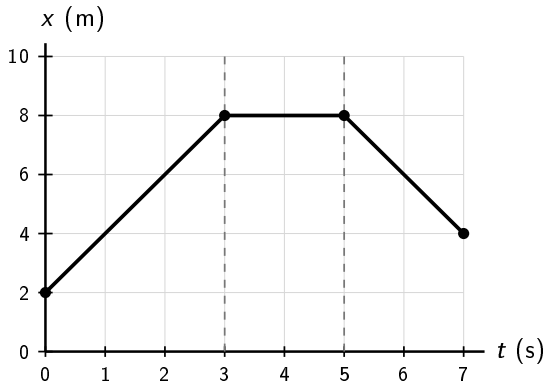
Slope on $x-t$ becomes height on $v-t$.

Class 4-2 companion document

01

START FROM THE SAME POSITION-TIME GRAPH

Do not invent new motion data. Reuse the exact four events from Class 4-1 and identify the three straight motion intervals.

**EXACT MOTION DATA**

(0 s, 2 m) $\rightarrow$ (3 s, 8 m)
 $\rightarrow$ (5 s, 8 m) $\rightarrow$ (7 s, 4 m)

THREE INTERVALS

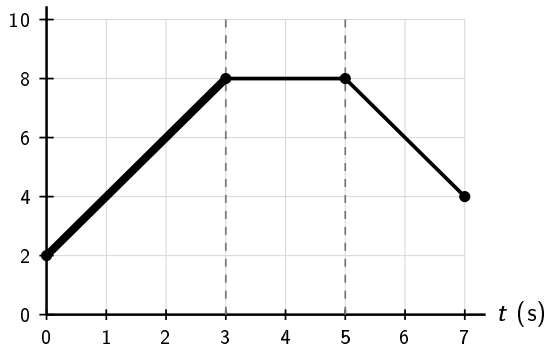
0 to 3 s: rising
3 to 5 s: horizontal
5 to 7 s: falling

$$v = \text{slope} = \frac{\Delta x}{\Delta t}$$

02

CALCULATE ONE SLOPE FOR EACH STRAIGHT SEGMENT

Velocity is not read from graph height. It is the slope: change in position divided by change in time.



INTERVAL 1:

$$0 \leq t < 3 \text{ s}$$

$$\Delta x = 8 - 2 = +6 \text{ m}$$

$$\Delta t = 3 - 0 = 3 \text{ s}$$

$$v = \frac{+6}{3} = +2 \text{ m/s}$$

VELOCITY LEVEL: +2 m/s

This becomes one horizontal segment on the velocity-time graph over 0 to 3 s.

02**CALCULATE ONE SLOPE FOR EACH STRAIGHT SEGMENT**

The horizontal segment has zero rise, therefore zero velocity even though the position is 8 m.

**INTERVAL 2:**

$$3 \leq t < 5 \text{ s}$$

$$\Delta x = 8 - 8 = 0 \text{ m}$$

$$\Delta t = 5 - 3 = 2 \text{ s}$$

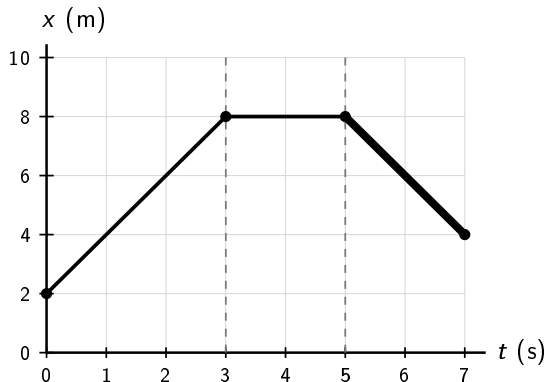
$$v = \frac{0}{2} = 0 \text{ m/s}$$

VELOCITY LEVEL: 0 m/s

Position is constant during this interval: the object is at rest.

02**CALCULATE ONE SLOPE FOR EACH STRAIGHT SEGMENT**

A falling position-time segment has negative slope, so the velocity is negative.

**INTERVAL 3:**

$$5 \leq t \leq 7 \text{ s}$$

$$\Delta x = 4 - 8 = -4 \text{ m}$$

$$\Delta t = 7 - 5 = 2 \text{ s}$$

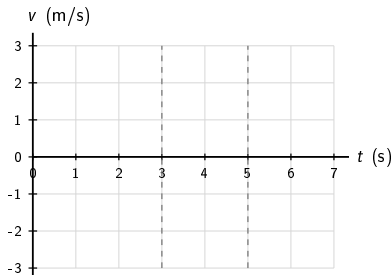
$$v = \frac{-4}{2} = -2 \text{ m/s}$$

VELOCITY LEVEL: -2 m/s

This becomes one horizontal segment below the time axis over 5 to 7 s.

03**YOUR TURN: CONSTRUCT THE VELOCITY-TIME GRAPH**

Before the answer appears, use the three velocities you calculated and place each one over its correct time interval.

**STUDENT TASK**

Draw one horizontal segment for each interval. Use the correct sign and keep the same time boundaries at $t = 3$ s and $t = 5$ s.

INTERVAL 1

0 to 3 s

$v = +2$ m/s

INTERVAL 2

3 to 5 s

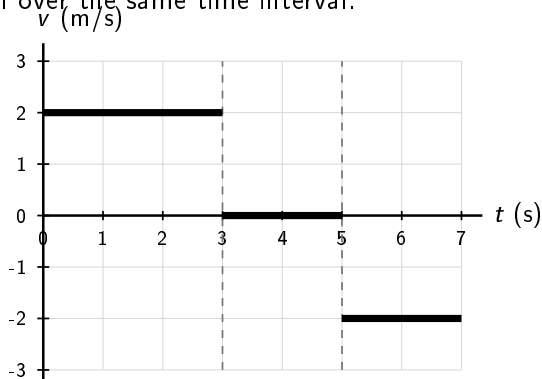
$v = 0$ m/s

INTERVAL 3

5 to 7 s

$v = -2$ m/s

Each constant slope on the position graph becomes a constant height on the velocity graph over the same time interval.



$$v(t) = \begin{cases} 2, & 0 \leq t < 3 \\ 0, & 3 \leq t < 5 \\ -2, & 5 \leq t \leq 7 \end{cases} \text{ m/s}$$

STEP 1

0 to 3 s: plot $v = +2$ m/s

STEP 2

3 to 5 s: plot $v = 0$ m/s

STEP 3

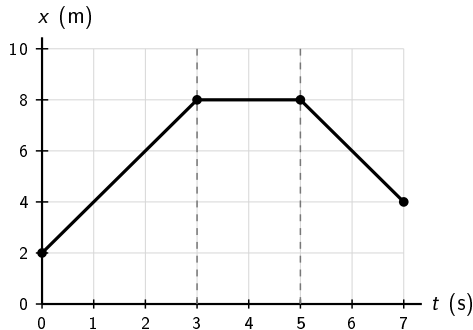
5 to 7 s: plot $v = -2$ m/s

AT $t = 3$ s AND $t = 5$ s

Velocity changes instantly at the idealized boundaries. Do not connect the levels vertically.

Look vertically at the same time interval: slope on x - t and height on v - t must describe the same motion.

POSITION-TIME



RIISING x - t

same time interval

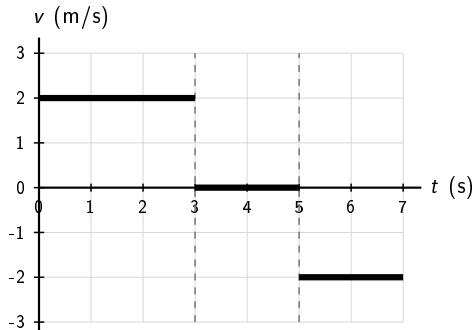
$$v = +2 \text{ m/s}$$

HORIZONTAL x - t

same time interval

$$v = 0 \text{ m/s}$$

VELOCITY-TIME



FALLING x - t

same time interval

$$v = -2 \text{ m/s}$$

A correct velocity graph must reproduce direction, rest, and the net displacement of the original motion.

0-3 s | MOVE RIGHT

x increases by 6 m

$$v = +2 \text{ m/s}$$

3-5 s | WAIT

x stays at 6 m

$$v = 0 \text{ m/s}$$

**5-7 s | RETURN
LEFT**

x decreases by 4 m

$$v = -2 \text{ m/s}$$

$$\Delta x = (+2)(3) + (0)(2) + (-2)(2)$$

$$\Delta x = 6 + 0 - 4 = +2 \text{ m}$$

$$x_f = x_i + \Delta x = 2 + 2 = 4 \text{ m}$$

The signed area under the v - t graph reproduces the final position of the original x - t graph.

07 THE FIVE-STEP x - t TO v - t METHOD

Use this exact sequence whenever a piecewise position-time graph is made of straight segments.

1 | SPLIT

Mark every straight time interval.

2 | SLOPE

Compute $\Delta x / \Delta t$.

3 | SIGN

Rising +, flat 0, falling -.

4 | HEIGHT

Plot that v level on the same interval.

5 | CHECK

Area under v - t must match Δx .

$$\begin{array}{ccccccc} (0, 2) & \rightarrow & (3, 8) & \rightarrow & (5, 8) & \rightarrow & \\ (7, 4) & \Rightarrow & v = +2, & 0, & -2 & \text{m/s} & \end{array}$$

SLOPE ON x - t = HEIGHT ON v - t

Construct. Calculate. Match the interval. Verify the motion.